



Instructor: Date:

Learner: Regional Branch:

Program: Center:

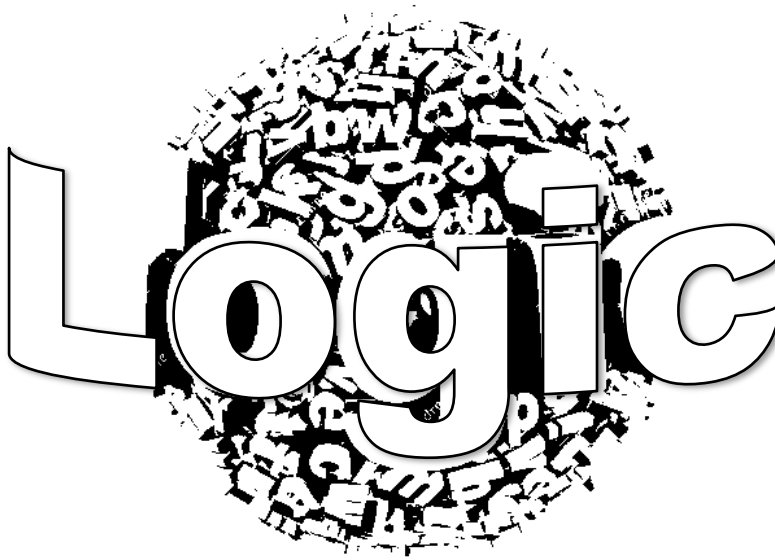
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Let's Start!

The Importance of Boolean Logic



Learning Outcome:

Exchange information about logic, its principles, and methods in simple spoken and written situations.

Ask your partner:

- Are you good at algebra?
- Do you think programmers have to be good at algebra? Why?

Can Do Statements:

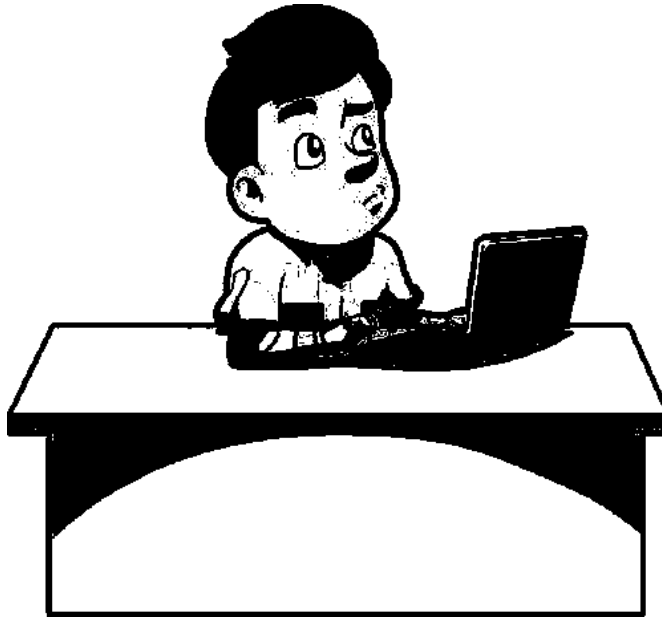
- I can understand written information about the concepts of Boolean logic and Boolean algebra.
- I can use the concept of two valued logic.
- I can write some sentences with examples of two valued logic.





Let's Explore!

Two Important Aspects to Take into Account

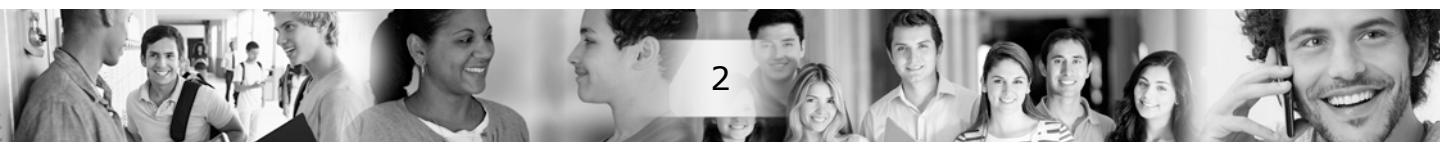


Instructions: Read the text below.

Many people don't get to understand a lot about two different but important programming concepts: Boolean logic and Boolean algebra. Don't get confused as they do not refer to the same element. First of all, the Boolean logic represents a mathematical expression in one out of two values: true or false. Plus, Boolean logic is the mathematical basis to understand the Boolean algebra. We use the Boolean logic in the computer science and it is fundamental to learn how to use formal or macro languages to program computers.

As the Boolean logic does not take two values but only one, we also name it the two-valued logic. When we use the two-valued logic, we create expressions, which are a combination of logical operands and operators. In simple words, operands are basically statements (which are true or false) and operators refer to the logical conventions AND, OR and NOT. Don't get confused! In arithmetic, we say that operands are numbers and operators are mathematical actions like addition, subtraction, multiplication, and division.

Many people don't get the idea of the two valued logic or the Boolean logic without examples, so let's check some. When we write $5 > 3$, we can say that the expression is true because 5 is more than 3. Another example is: I am tall AND I am the owner of Microsoft. In this case, the Boolean logic tells us that the expression is false because I can be tall, but it is not possible that I become the owner of Microsoft.





Let's Practice!

Understanding concepts about Boolean

Instructions: Proofread the sentences. Correct the mistakes.

1. The Boolean logic has two components: true or false.

2. Boolean logic and Boolean algebra are simple or confusing.

3. The Boolean algebra by logic use operands.



Let's Practice!

Two valued logic

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

Correct Information about Two Valued Logic	Incorrect Information about Two Valued Logic

1. We use True or False statements.
2. We take two values at a time.
3. We mix connectors with symbols.
4. We use operands and operators.
5. We don't use yes or no.
6. We don't create expressions.



Extension Activity

Talking about Boolean Language

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about types of Boolean
- Describe how they work.
- Write a 5 sentences about types of Boolean language.
- Show your sentences to your classmates and teacher.

Self-Assessment

Yes

No

- I deepen my knowledge on Boolean language.
- I express information through sentences.
- I show information to other people.

☐	☐
☐	☐
☐	☐



Target Words

- **value** (noun): a quantity represented by a symbol.
e.g. it states that something can have two values: it can be true or false
- **expression** (noun): numbers, symbols, operands and operators grouped together that show the value of something.
e.g. Moving on, an expression is the grouping of logical operands and operators
- **basis** (noun): a pillar, the element that permit other component exist
e.g. Plus, Boolean logic is the mathematical basis to understand the Boolean algebra.
- **operand** (noun): a quantity used for a mathematical operation.
e.g. In Boolean logic an operand is a statement (true or false)
- **operator** (verb): a symbol to express a mathematical operation.
e.g. ... and operators are logical connectors AND, OR and NOT.



Learning Points

Simple Connectors

You use **connectors** to join or link words or sentences. They are of great importance to create coherence and cohesion when talking or writing.

The most common **connectors** that we use in English are **AND**, **OR** and **BUT**. Let's check how they work:

We use the connector **AND** to join two sentences that are grammatically similar or when we list different elements.

On the other hand, we use the connector **OR** to express an alternative or an option.

Finally, we use the connector **BUT** to contrast ideas.

For example:

*Juan **and** Carlos are experts on systems theory.*

*We can use the name "systems theory" **or** "systems science". You decide which one you want to use.*

*Carla wants to work with algebra **but** she is bad at it.*

Examples

- *When we use the two valued logic, we create expressions, which are a combination of logical operands and operators.*
- *In simple words, operands are basically statements (which are true or false).*
- *As the Boolean logic does not take two values but only one, we also name it the two valued logic.*



Learning Points

Present Simple Negative

We use the **present simple** to talk about habits or facts.

We can use the **present simple** in both, its affirmative or negative form. When we use it in its **negative** form, we refer to elements that are not facts or don't make part of our routines. The structure that we use to write negative sentences using this tense is:

Subject + auxiliary verb "do" or "does"* + not + verb in base form + complement.

Don't forget that we use the auxiliary verb "**do**" for the pronouns *I, you, we, and they*. On the other hand, we use the auxiliary verb "**does**" for the pronouns *he, she, and it*.

Examples:

*I **don't understand** anything about algebra.*

*She **does not study** at home, that is why she is so bad with numbers.*

We can also use imperatives, orders or commands, to express ideas in the negative form of the present simple tense. For this, we omit the pronoun and start the sentence with the verb.

For example:

***Don't* forget** to study Boolean logic for the exam.*

***Do not come** until you understand what valued logic is.*

*It is important to know that when we use imperatives, it is not common to use the auxiliary verb does.

Examples

- *Many people don't get to understand a lot about two different but important programming concepts: Boolean logic and Boolean algebra.*
- *Many people don't get the idea of the two valued logic or the Boolean logic without examples, so let's check some.*